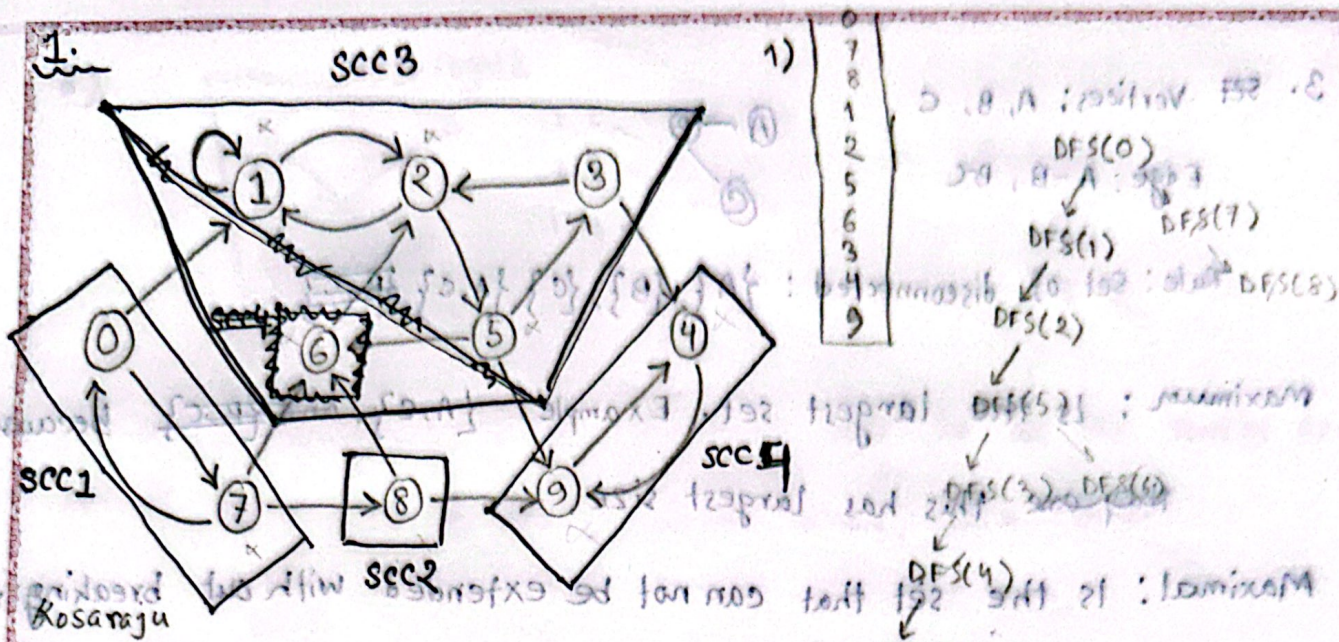


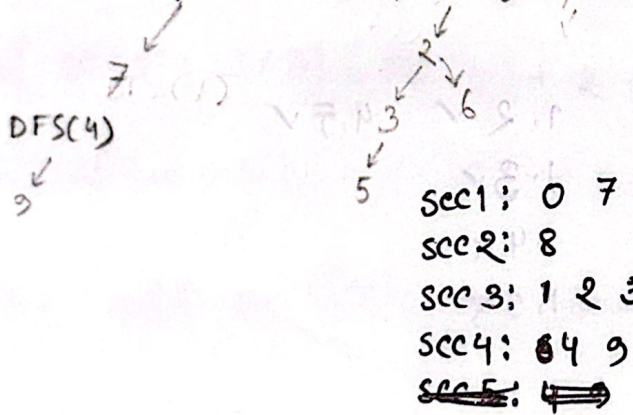


1. Sort edges according to finishing time

2. Transpose Graph

3. DFS

3) DFS(0) DFS(8) DFS(1) DFS(6)



Total SCC = 4

2. i

Transpose Graph ensures that the traversal does not happen in other SCC. Even after traversing edges $u \rightarrow v$ and $v \rightarrow u$ can be reachable in own SCC. But reversing isolate from going to another SCC.

3. Vertices: A, B, C

Edge: A-B, BC



Rule: Set of disconnected: $\{A\}, \{B\}, \{C\}, \{A, C\}, \{B, C\}$

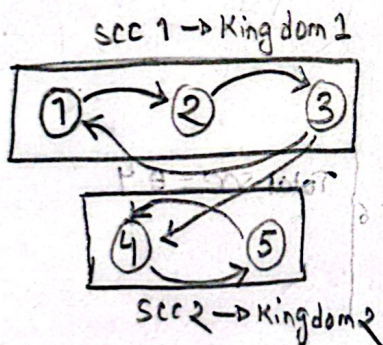
Maximum: Is the largest set. Example: $\{A, C\}$ and $\{B, C\}$ because they are this has largest size.

Maximal: Is the set that can not be extended without breaking rule.

Example: $\{B\}$ is maximal cause cannot add A or C.

$\{A, C\}$ is also maximal.

4. i) Planets and Kingdoms



1, 2 ✓ 4, 5 ✓

1, 3 ✓

1, 4 ✗

1, 5 ✗

2, 4 ✗

2, 5 ✗

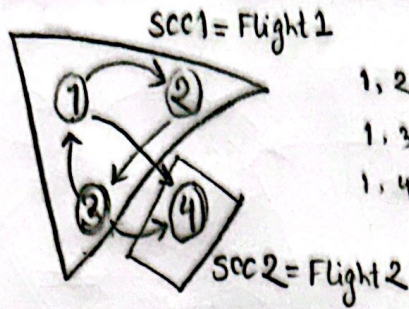
SCC1: $\{1, 2, 3\} \rightarrow \text{label: } 1$

output:

SCC2: $\{4, 5\} \rightarrow \text{label: } 2$

Here SCC Represents Kingdoms. So, with Kosaraju we can tell which planet belongs to which kingdom, and which are reachable in $u \rightarrow v$ and $v \rightarrow u$ manner.

ii)



1, 2 ✓ 2, 3 ✓
1, 3 ✓ 2, 4 x
1, 4 ✓

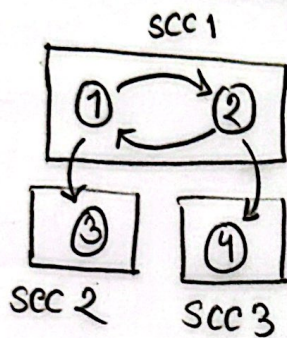
Output

No

4 2

As the entire graph is not an entire SCC So all the routes are not possible. 2 and 4 do not follow the $u \rightarrow v, v \rightarrow u$ manner.

iii)



Room No. Coin

1 4

2 5

3 2

4 7

Output

16

$$\text{Coin}(\text{SCC1} + \text{SCC2}) = (4 + 5) + 2 = 11$$

$$\text{Coin}(\text{SCC1} + \text{SCC3}) = (4 + 5) + 7 = 16 \checkmark$$

Using DAG on SCC's can count total collected coin.